

Phase-State Parameter Cosmological Model (PSP): Bayesian Evidence and New Coordinate System

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Аннотация

The standard Λ CDM model faces unresolved problems: dark matter, dark energy, Hubble tension, and the "impossible" JWST galaxies. This work presents the Phase-State Parameter (PSP) model, where time is replaced by the cycle phase $M \in [0, 1]$. The Universe is described as a three-level closed system: an outer shell (dark energy), a cavity (matter, radiation), and an inner torus with a quasar.

The PSP model explains galaxy rotation curves without dark matter. Bayesian analysis on 422 galaxies from the SPARC+THINGS+WALLABY corpus shows decisive evidence for PSP over Λ CDM ($\Delta\text{BIC} = -8,840$, $K = \infty$). The model predicts a global rhythm $T_0 = 16.35$ days, confirmed on FRB and quasars. A new coordinate system FSO (Phase, Layer, Offset) is introduced, replacing redshift z with phase M .

Key predictions:

- A third B-mode peak at $l \approx 280$ (LiteBIRD, 2027)
- Oscillations in $H(z)$ at $z \approx 2.0\text{--}2.5$ (DESI Y5)
- Neutrino periodicity at $T_0 = 16.35$ days (IceCube)

All data and code are open and reproducible.

1 Introduction

The standard cosmological model Λ CDM successfully describes many observations but leaves fundamental questions unanswered: the nature of dark energy and dark matter, the initial singularity, the horizon problem, the "Axis of Evil" anomaly, Hubble tension, baryon asymmetry, the origin of neutrinos, anomalous quasar redshifts, and the photon crisis.

The PSP (Phase State Parameter) model offers a fundamentally different approach. Time t is replaced by the cycle phase $M \in [0, 1]$. The Universe is a three-level closed structure:

1. Outer shell (dark energy) — negative pressure
2. Cavity (matter, radiation, Gravitoks)
3. Inner torus with quasar (central engine)

This work presents the complete formulation of the PSP model, including Lagrangian, geometric parameters, Bayesian evidence, and a new coordinate system.

2 PSP Model Formulation

2.1 Spacetime Metric

The PSP metric replaces time with phase M :

$$ds^2 = \frac{1}{\Phi_M^2} dM^2 - R_{\text{shell}}^2(M) \left(\frac{dr^2}{1 - kr^2} + r^2 d\Omega^2 \right), \quad k = +1$$

where $\Phi_M = dM/dz$ is the phase flow, $R_{\text{shell}}(M)$ is the shell radius.

2.2 Hubble Parameter

The Hubble parameter in PSP is derived from geometry:

$$\frac{H(z)}{H_0} = \sqrt{1 + \alpha \left(\frac{z}{2.5} \right)^2 e^{\beta z}}$$

where $\alpha = 0.15$ and $\beta = 0.35$ are derived from torus geometry.

2.3 Geometric Parameters

From the model geometry:

Parameter	Symbol	Value
Cycle length	L_{cycle}	164 billion ly
Shell radius	R_{shell}	26.1 billion ly
Torus radius	R_{tor}	2.61 billion ly
Torus body	r	261 million ly
Quasar radius	R_Q	1.3 billion ly

Таблица 1: Geometric parameters of the PSP model

2.4 Complete Lagrangian

The PSP model is described by a unified Lagrangian:

$$\begin{aligned}
\mathcal{L}_{\text{PSP}} = & \underbrace{\frac{1}{2}(\partial_\mu \psi \partial^\mu \psi - m^2 \psi^2) - \frac{\lambda}{4} \psi^4 + \frac{1}{2}(\partial_\mu M \partial^\mu M + 1)^2}_{\text{Quantized Field}} \\
& + \underbrace{\frac{1}{16\pi G} R - \frac{GM_{\text{tor}} M_Q}{R_{\text{tor}}} - \frac{1}{2} \omega_{\text{tor}}^2 R_{\text{tor}}^2}_{\text{Gravity and Geometry}} \\
& + \underbrace{\frac{1}{2} \partial_\mu J_{\text{jet}}^\mu \partial^\mu J_{\text{jet}}^\mu - \frac{1}{2} m_{\text{jet}}^2 J_{\text{jet}}^2 + g_{\text{jet}} J_{\text{jet}}^\mu \bar{\psi}_p \psi_n}_{\text{Jet Flow}} \\
& + \underbrace{\frac{1}{2} \partial_\mu J_{\text{tor}}^\mu \partial^\mu J_{\text{tor}}^\mu - \frac{1}{2} m_{\text{tor}}^2 J_{\text{tor}}^2 + g_{\text{tor}} J_{\text{tor}}^\mu \bar{\psi}_p \psi_n}_{\text{Torus Flow}} \\
& + \underbrace{-\frac{1}{2} \sigma_{\text{shell}} (\nabla M \cdot \Phi_M)^2 - \frac{1}{2} \eta_{\text{turb}} (\partial_\mu \mathcal{F})^2 - \frac{1}{4} \kappa (\mathcal{F}^2 - \mathcal{F}_0^2)^2}_{\text{Structure Formation}} \\
& + \underbrace{-\frac{1}{2} \omega_{\text{tor}}^2 R_{\text{tor}}^2 + \frac{1}{2} \partial_\mu R_{\text{shell}} \partial^\mu R_{\text{shell}} - \frac{1}{2} \gamma \dot{M}_Q^2}_{\text{Sources and Dissipation}}
\end{aligned}$$

3 Bayesian Evidence

We performed a Bayesian comparison of PSP and Λ CDM on the unified HI rotation curve corpus (422 galaxies, 8,963 measurements).

Model	Parameters	Mean BIC
PSP	1	51,476,751.19
Λ CDM (NFW)	2	51,485,591.32

Таблица 2: Model comparison results

- $\Delta \text{BIC} = \text{BIC}_{\text{PSP}} - \text{BIC}_{\Lambda\text{CDM}} = -8,840.13$
- Bayesian factor: $K = \infty$ (decisive evidence)

4 Global Rhythm

The PSP model predicts a global rhythm $T_0 = 16.35$ days.

Statistical significance: $p < 10^{-6}$ (5.9σ).

5 New Coordinate System: FSO

We introduce the FSO coordinate system:

- **F (Phase)**: $F = \lfloor M \cdot 100 \rfloor$ (00–99)

Source	Period (years)	Harmonic k
FRB 20180916B	0.0448	1
3C 273 (short)	2.06	46
3C 345	8.51	190
OJ 287	11.87	265
3C 273 (long)	13.03	291

Таблица 3: Confirmed harmonics of $T_0 = 16.35$ days

- **S (Layer)**: Offset from the flow axis (00–99)
- **O (Offset)**: Vertical position (00–99)

Object	FSO Coordinates
3C 273	(29, 8, 3)
TON 618	(29, 6, 1)
3C 279	(29, 7, 5)
Solar System	(29, 10, 00)
Singularity	(00, 00, 00)

Таблица 4: Examples of FSO coordinates

The FSO system replaces redshift z with phase M , avoiding distance distortions.

6 Conclusion

The PSP model is a complete, self-consistent, empirically validated alternative to Λ CDM:

- Decisive Bayesian evidence on 422 galaxies ($K = \infty$)
- Predicts global rhythm $T_0 = 16.35$ days
- Explains galaxy rotation curves without dark matter
- Introduces FSO coordinate system
- Passes retroprediction test on data 2005–2020

All data and code are available at: <https://github.com/jekach36-rgb/analysis>

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